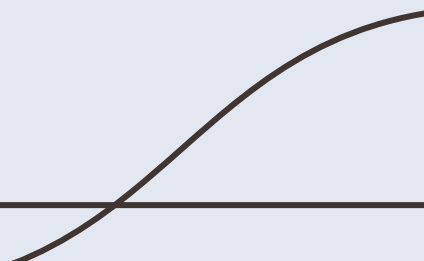


How to detect a GWB The Frequentist Way

Kyle Gersbach - VIPER PTA Summer School
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The Plan

- A reminder of frequentist statistics
- Maximizing the likelihood
- The PTA optimal statistic (OS)
 - As a detection statistic
 - As a parameter estimator
- Extensions of the optimal statistic
 - Pair Covariance
 - Per-Frequency
 - Multi-Component

What is a frequentist statistic?

How frequently do events happen?

- The key is that it assumes many events happen!

of Sigma $\rightarrow$ frequentist

What is a statistic? — Just a number!

- The importance of a statistic is its utility
- Make sure that it does what you want it to do

What makes a static ‘optimal’?

If we have an estimator for the GWB Amplitude

- $\hat{A}^2$

The variance on that estimator is (i.e. spread on the fit)

- $(\sigma_{\hat{A}^2})^2 = \langle (\hat{A}^2)^2 \rangle - \langle \hat{A}^2 \rangle^2$

An ‘optimal’ estimator is one which minimizes its own variance!

From the PTA likelihood

$$\ln p(\delta t | A_{\text{gw}}^2) = -\ln |\pi C(A_{\text{gw}}^2)| - \delta t^T C^{-1}(A_{\text{gw}}^2) \delta t$$

Taylor series expansion (assume small A_{gw})

$$\ln p(\delta t | A_{\text{gw}}^2) \approx -\sum_a \left(\text{tr}(\ln P_a) + \delta t_a^T P_a^{-1} \delta t_a \right) - \sum_{ab} \delta t_a^T P_a^{-1} S_{ab} P_b^{-1} \delta t_b$$

Maximize the likelihood ratio for A_{gw}

$$\ln \Lambda = \ln p(\delta t | A_{\text{gw}}^2) - \ln p(\delta t | A_{\text{gw}}^2 = 0)$$

$$\frac{d \ln \Lambda}{d A_{\text{gw}}^2} = \sum_{ab} \delta t_a^T P_a^{-1} S_{ab} P_b^{-1} \delta t_b$$

The PTA (original) optimal statistic (OS)

$$\hat{A}^2 = \frac{\sum_{a < b} \delta t_a^T P_a^{-1} \tilde{S}_{ab} P_b^{-1} \delta t_b}{\sum_{a < b} \text{tr} \left(P_a^{-1} \tilde{S}_{ab} P_b^{-1} \tilde{S}_{ba} \right)}$$

$$\sigma_{\hat{A}^2} = \left(\sum_{a < b} \text{tr} \left(P_a^{-1} \tilde{S}_{ab} P_b^{-1} \tilde{S}_{ba} \right) \right)^{-1/2}$$

$$P_a = \langle \delta t_a \delta t_a^T \rangle$$

$$S_{ab} = \langle \delta t_a \delta t_a^T \rangle$$

$$S_{ab} = A_{\text{gw}}^2 \tilde{S}_{ab}$$

Contains Hellings
and Downs!

[Anholm et al 2009], [Chamberlin et al 2015]

The better OS

Pair correlated
power

$$\rho_{ab} = \frac{\mathbf{X}_a^T \hat{\phi} \mathbf{X}_b}{\text{tr} \left[\mathbf{Z}_b \hat{\phi} \mathbf{Z}_a \hat{\phi} \right]},$$
$$\sigma_{ab,0} = \text{tr} \left[\mathbf{Z}_b \hat{\phi} \mathbf{Z}_a \hat{\phi} \right]^{-1/2}$$

$$\hat{A}_{\text{gw}}^2 = \frac{\sum_{a < b} \rho_{ab} \Gamma_{ab} / \sigma_{ab,0}^2}{\sum_{a < b} \Gamma_{ab}^2 / \sigma_{ab,0}^2}.$$
$$\sigma_{\hat{A}^2,0} = \left(\sum_{a < b} \frac{\Gamma_{ab}^2}{\sigma_{ab,0}^2} \right)^{-1/2}.$$

unweighted Fourier
coefficients

$$\mathbf{X}_a = \mathbf{F}_a^T \mathbf{P}_a^{-1} \delta \mathbf{t}_a,$$

Covariance of X

$$\mathbf{Z}_a = \mathbf{F}_a^T \mathbf{P}_a^{-1} \mathbf{F}_a.$$

$\hat{\phi}$ Is a unit-power-law
spectrum

[Pol, Taylor, Romano 2022], [Gersbach et al 2025]

Why's it better?

Working in fourier domain instead of time domain

- Matrices shrink! ($N_{\text{toa}} \times N_{\text{toa}}$) $\rightarrow$ ($2N_{\text{freq}} \times 2N_{\text{freq}}$)
- Inversions are more stable!

Pairwise values are handy with visualization!

- In this form we linearly fit for an HD curve

What about those matrix products?

We need to calculate our X and Z matrices:

$$\begin{aligned}\mathbf{X}_a &= \mathbf{F}_a^T \mathbf{P}_a^{-1} \delta \mathbf{t}_a, \\ \mathbf{Z}_a &= \mathbf{F}_a^T \mathbf{P}_a^{-1} \mathbf{F}_a.\end{aligned}$$

The problem is, the autocorrelation terms ($\mathbf{P}_a$) need estimates of the total power. So how do we get these?

A Bayesian CURN Search

Why don't we use a Bayesian search!

Instead of searching for full correlations

- [used to be] VERY SLOW

Use a Common Uncorrelated Red Noise (CURN) search!

- Accurate estimates for everything you need
- At a rate $>100x$ faster than a correlated search

An additional snag

We have distributions of posteriors, what now?

Which vector of parameters do we choose?

1. Select the maximum likelihood value
 - a. Biased in PTAs with significant intrinsic red noise
2. Marginalize over many parameter vectors!
 - a. Don't choose one, choose a representative set of them!
 - b. Many random draws = True posterior

[Vigeland et al 2018]

The Noise Marginalized OS (NMOS)

Use 1,000 - 10,000 random samples from the CURN search

- Run each through the OS
- Get a distribution of means, uncertainties, and SNRs

NOTE: Be careful, the distribution on $\hat{A}_{\text{gw}}^2$ is NOT the full distribution on A^2 , its only the distribution of means!

- Can use Uncertainty Sampling to get full amplitude distribution

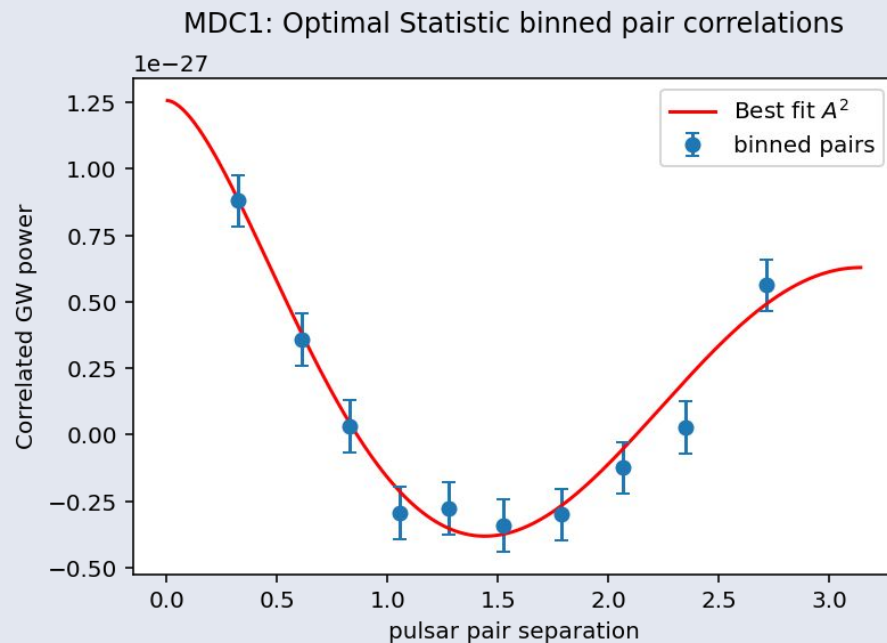
[Gersbach et al 2025]

What does the OS actually do?

Fitting data to a model

The 'data' is the pairwise correlated estimators

The 'model' is the amplitude modulated HD curve



The OS as a detection statistic

SNR is approximate # of sigma

- Monotonic function
- SNR distribution $\neq$ Gaussian

MUST calibrate SNR $\rightarrow$ True sigma

- Destroy correlations with phase shifts [[Taylor et al 2016](#)]
- Use the GX2 to analytically estimate [[Hazboun et al 2023](#)]

$$\text{SNR} = \frac{\hat{A}_{\text{gw}}^2}{\sigma_{\hat{A}^2,0}}.$$

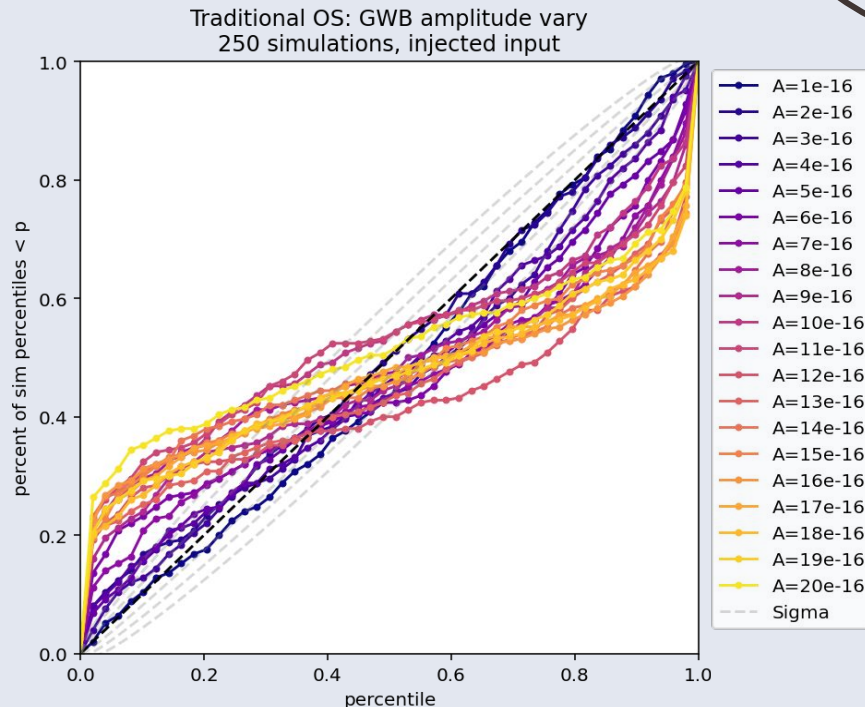
The OS as a parameter estimator

Fairly accurate in the weak-signal regimes

- $\text{SNR} < \sim 2-3$

A_{gw} estimator wrong for loud GWBs

- Independent pair assumption is bad!



PP plots with increasing amplitude
(more diagonal is better)

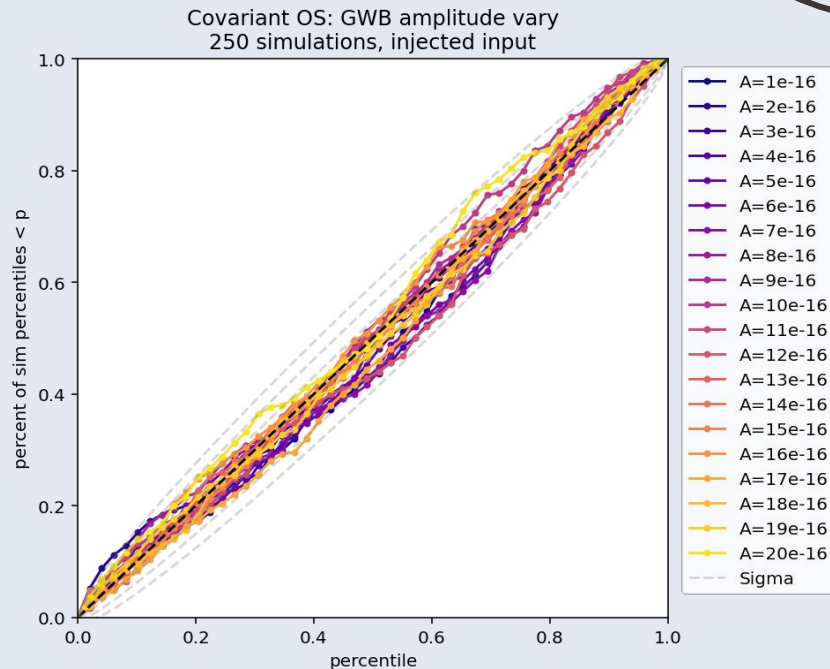
The first extension: Pair covariance

Enable correlations between pulsar pairs

- More compute
- MUCH more accurate

Can be combined with any other OS extension!

[Allen, Romano 2023] [Gersbach et al 2025]



PP plots with increasing amplitude
(more diagonal is better)

The second extension: Per-Frequency

Generalizing OS for any spectral shape

- Correlations per-frequency

NEED Pair covariance

- Accuracy matters!

[Gersbach et al 2025]

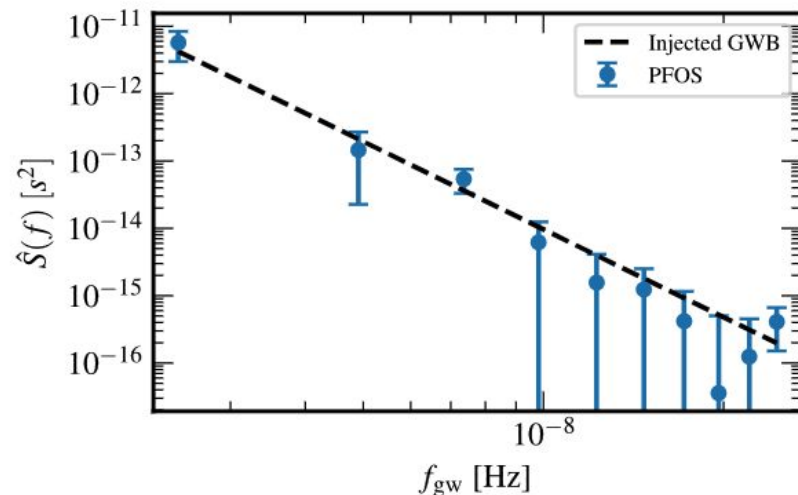


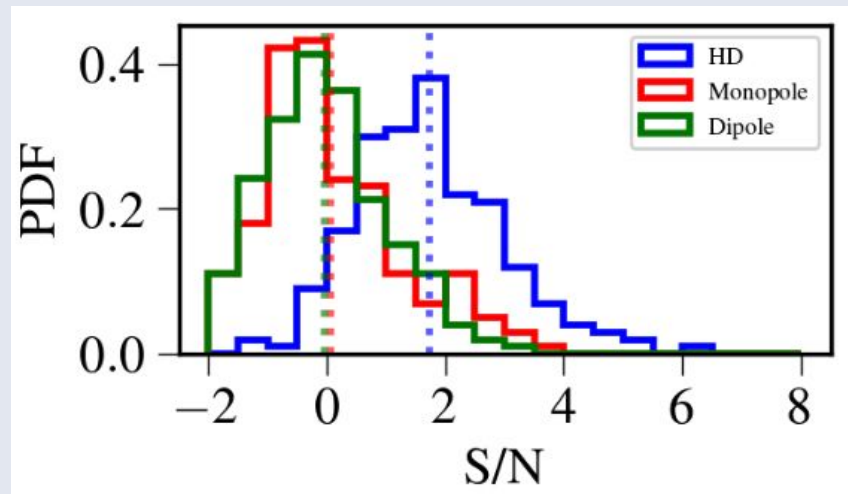
FIG. 3. The recovered pulsar-pair correlated GWB spectrum from the (broadband-normalized) PFOS given the injected simulation parameters with point estimates on $S(f_k)$ and associated 1-sigma error bars. The black dashed line indicates the injected GWB PSD.

The third extension: Multi-component

Generalizing OS for any correlation function simultaneously

- HD + Monopole + Dipole

Significance of noise processes in the presence of GWB



The SNR distribution for each process evaluated simultaneously

[Sardesai et al 2023]

[Sardesai et al 2023]

All of this + more in DEFIANT!

Comprehensive OS python package built by me!

- ~2 orders of magnitude faster than enterprise_extensions
- Support for all extensions and their combinations!
- Numerical null distributions
- HEAVILY documented